Ex 11 $f: \mathbb{R}^d \to \mathbb{R}$ and f is differentiable ($f \in \mathcal{D}(\Omega)$)

then for $\forall\, x, y, z \in \mathbb{R}^d$

$$D_f(x,z) = D_f(x,y) + \langle x-y ; \nabla f(y) - f(z) \rangle + D_f(y,z) \quad (a)$$

This is the same as

$$f(x) - f(z) - \langle \nabla f(z) ; x-z \rangle = f(x) - f(y) - \langle \nabla f(y) ; y-x \rangle$$

$$+ \langle z-y ; \nabla f(z) \rangle - f(z) + f(y) + \langle \nabla f(y) ; y-x \rangle +$$

cancel the common terms on both sides ($f(y)$)

$$- \langle \nabla f(z) ; x-z \rangle = - \langle x-y ; \nabla f(y) \rangle + \langle z-y ; \nabla f(z) \rangle + f(y) - f(z) + \langle \nabla f(y) ; y-x \rangle$$

also $\langle x-y ; \nabla f(y) - f(z) \rangle = - \langle \nabla f(y) ; y-x \rangle + \langle z-y ; \nabla f(z) \rangle$

as a result this is the same as

$$- \langle \nabla f(z) ; x-z \rangle = - \langle \nabla f(z) ; z-y \rangle - \langle y-x ; \nabla f(z) \rangle \quad \text{the same as}$$

$$0 = \langle 0 ; \nabla f(z) \rangle = \langle x-y+y-z+z-x ; \nabla f(z) \rangle$$

(i) If $x = z$ then $D_f(x,z) = f(x) - f(x) - \langle \nabla f(x) ; x-x \rangle = 0$

As a result we get

$$D_f(x,y) + D_f(y,x) = - \langle x-y ; \nabla f(y) - \nabla f(x) \rangle = \langle y-x ; \nabla f(x) - \nabla f(y) \rangle$$

(ii) If $f(x) = \frac{1}{2}\|x\|^2$

$\nabla f(v) = v$ and $D_f(x,y) = f(x) - f(y) - \langle \nabla f(y); x-y \rangle =$

$= \|x\|^2 - \|y\|^2 - \langle y; x-y \rangle =$

$= \|x\|^2 - \|y\|^2 - \langle y; x \rangle + \|y\|^2 =$

$\|y\|^2 - \langle x; y \rangle - \|x\|^2 + \langle x; y \rangle + \|y\|^2$ and $\frac{1}{2}\|x-y\|^2 = \langle x-y; x-y \rangle \frac{1}{2} =$

As a result

$\langle x-y; \nabla f(y) - \nabla f(z) \rangle = \langle z-y; y-x \rangle$

Also

$D_f(x,z) - D_f(x,y) - D_f(y,z) = \|x\|^2 - \|y\|^2 - \|z\|^2 + \langle x; y \rangle + \langle x; z \rangle$

$= \underbrace{\frac{1}{2}\|x\|^2 + \frac{1}{2}\|z\|^2 - \langle z; x \rangle}_{\frac{1}{2}\|z-x\|^2} + \langle x; y \rangle + \|y\|^2 - \frac{1}{2}\|x\|^2 =$

(ii) If $f(x) = \frac{1}{2}\|x\|^2$ then

$\nabla f(v) = v \iff \langle z-y; y-x \rangle = \langle z; \nabla f(y) - \nabla f(x) \rangle$

Also $D_f(x,y) = \|x\|^2 - \|y\|^2 - \langle x-y; y \rangle = \langle x-y; x \rangle - \langle y; x-y \rangle$

$= \langle x-y; x \rangle - \langle y; x-y \rangle = \langle x-y; x-y \rangle =$

$= \|x-y\|^2 = D_f(y,x) \iff \|x-y\|^2 = \langle x-y; x-y \rangle =$

As a result

(a) $\iff \left(\|z-x\|^2 = \frac{1}{2}\|y-x\|^2 + \langle z-y; y-x \rangle + \frac{1}{2}\|z-y\|^2 \right)$ when

$f(x) = \frac{1}{2}\|x\|^2$.

$\frac{1}{2}\|x-z\|^2 = \frac{1}{2}\|x\|^2 \frac{1}{2}\langle x-z; x-z\rangle = \frac{1}{2}\|x\|^2 + \frac{1}{2}\|z\|^2 - \langle x; z\rangle$

Ex 12 If $f: \mathbb{R}^d \to \mathbb{R}$ be $f \in C^2(\mathbb{R}^d)$: Then

If f is convex if and only if $\nabla^2 f(x) \geq 0$:

1) $\langle \nabla f(x) - \nabla f(y); x-y\rangle = \int_0^1 \nabla^2 f(y+t(x-y))(x-y); x-y\rangle dt$

$v := x-y$ we get $\int_0^1 \langle \nabla^2 f(y+vt)\cdot v; v\rangle dt$

And because $\nabla^2 f(x) \geq 0$ then $\langle \nabla^2 f(y+vt) v; v\rangle \geq 0$

as a result $\langle \nabla f(x) - \nabla f(y); x-y\rangle \geq 0$ ///

2) If f is convex then

$f(x) \geq f(y) + \langle \nabla f(y); x-y\rangle$ define $x := y + hv$ $h\in\mathbb{R}$, $v\in\mathbb{R}^d$

then $f(y+hv) \geq f(y) + h\langle \nabla f(y); v\rangle$ also because $f \in C^2(\mathbb{R}^d)$

$f(y+hv) = f(y) + h\langle \nabla f(y); v\rangle + h^2 \frac{\langle \nabla f(y)\cdot v; v\rangle}{2} + o(\|hv\|^2)$

As a result $\frac{h^2}{2}\langle \nabla f(y) v; v\rangle + o(\|hv\|^2) \geq 0 \quad \forall h, v, y$

when $h \to 0$ $\quad \langle \nabla f(y) v; v\rangle \geq 0 \iff \nabla f(y) \geq 0 \quad \forall y$:

Ex 13 $f: \mathbb{R}^d \to \mathbb{R}$ is convex, $x_i \in \mathbb{R}^d$, $\lambda_i \in \mathbb{R}^+$ and $\sum \lambda_i = 1$ $\forall i = 1..m$

Then $f\left(\sum_{i=1}^{m} \lambda_i x_i\right) \le \sum_{i=1}^{m} \lambda_i f(x_i)$

1) For $m = 2$ $\lambda_1 := \lambda$ then $\lambda_2 = 1 - \lambda$

$f(\lambda x_1 + (1-\lambda) x_2) \le \lambda f(x_1) + (1-\lambda) f(x_2)$ // by convexity

2) If $m = k$ is true then show for $m = k+1$

$\lambda'_k := \lambda_k + \lambda_{k+1}$ and $x'_k = \dfrac{\lambda_k x_k + \lambda_{k+1} x_{k+1}}{\lambda_k + \lambda_{k+1}}$ then

$\lambda'_i := \lambda_i$ and $x'_i := x_i$ $\forall i = 1, 2, \dots, k-1$

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) = f\left(\sum_{i=1}^{k} \lambda'_i x'_i\right) \le \sum_{i=1}^{k} \lambda'_i f(x'_i) =$$

$$= \sum_{i=1}^{k-1} \lambda_i f(x_i) + \left(\lambda_k + \lambda_{k+1}\right) \cdot f\left(\frac{\lambda_k x_k + \lambda_{k+1} x_{k+1}}{\lambda_k + \lambda_{k+1}}\right) \qquad (1)$$

$$f\left(\frac{\lambda_k}{\lambda_k + \lambda_{k+1}} x_k + \frac{\lambda_{k+1}}{\lambda_k + \lambda_{k+1}} x_{k+1}\right) \le \frac{\lambda_k}{\lambda_k + \lambda_{k+1}} f(x_k) + \frac{\lambda_{k+1}}{\lambda_k + \lambda_{k+1}} f(x_{k+1}) \quad (2) \text{ by def of convexity}$$

From (1) and (2)

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \le \sum_{i=1}^{k-1} \lambda_i f(x_i) + \lambda_k f(x_k) + \lambda_{k+1} f(x_{k+1}) =$$

$$= \sum_{i=1}^{k+1} \lambda_i f(x_i)$$ //

Ex 14 Let $f:\mathbb{R}^d \to \mathbb{R}$ and $f \in C^1(\mathbb{R}^d)$. Then the following statements are equivalent.

(i) f is μ-convex

(ii) $\mu\|x-y\|^2 \le 2D_f(x,y) \quad \forall\, x,y \in \mathbb{R}^d$

(iii) $\mu\|x-y\|^2 \le \langle \nabla f(x) - f(y); x-y\rangle \quad \forall\, x,y \in \mathbb{R}^d$

(i) $\Rightarrow$ (ii)

$g(x) = f(x) - \frac{\mu}{2}\|x\|^2$ then $g(x)$ is convex as a result

& $D_g(x,y) = g(x) - g(y) - \langle \nabla g(y); x-y\rangle \ge 0$ which is the same

$$f(x) - f(y) - \langle \nabla f(y); x-y\rangle + \left(\frac{\|x\|^2}{2} - \frac{\|y\|^2}{2} + \langle y; x-y\rangle\right)\cdot \mu \ge 0$$

on the other hand $\frac{\|x\|^2}{2} - \frac{\|y\|^2}{2} + \langle y; x-y\rangle = \frac{1}{2}\|x-y\|^2$

And finally we get

$$f(x) - f(y) - \langle \nabla f(y); x-y\rangle \ge \frac{\mu}{2}\|x-y\|^2 \quad //$$

(ii) $\Rightarrow$ (iii)

$$f(x) - f(y) - \langle \nabla f(y); x-y\rangle \ge \frac{\mu}{2}\|x-y\|^2$$

$+$

$$f(y) - f(x) - \langle \nabla f(x); y-x\rangle \ge \frac{\mu}{2}\|x-y\|^2$$

$=$

$$-\langle \nabla f(y), x-y\rangle + \langle \nabla f(x); x-y\rangle \ge \mu\|x-y\|^2$$ which is the same as

$$\langle \nabla f(x) - \nabla f(y); x-y\rangle \ge \mu\|x-y\|^2 \quad //$$

(iii) => (i)

$\langle \nabla f(x) - \nabla f(y), x-y \rangle \geq \mu \|x-y\|^2 \quad \forall x, y \in \mathbb{R}^d$

If $g(x)$ is convex then and $\langle \nabla g(x) - g(y), x-y \geq 0$ then

$$g(y) = g(x) + \int_0^1 \langle \nabla g(x + t(y-x)), y-x \rangle dt$$

$u := x + t(y-x)$ we get

$\langle \nabla g(u), \nabla g(x); u-x \rangle \geq 0$

Expanding

$\langle \nabla g(u), u-x \rangle \geq \langle \nabla g(x); u-x \rangle \; (:t)$

$\langle \nabla g(u); y-x \rangle \geq \langle \nabla g(x); y-x \rangle \; (\int_0^1)$

$\int_0^1 \langle \nabla g(u); y-x \rangle dt \geq \int_0^1 \langle \nabla g(x + t(y-x)), y-x \rangle dt \geq$

$\int_0^1 \langle \nabla g(x); y-x \rangle dt = \langle \nabla g(x); y-x \rangle$ or a result

$g(y) - g(x) \geq \langle \nabla g(x); y-x \rangle$ Now let g $g(x) = f(x) - \frac{\mu}{2}\|x\|^2$

then we get

$\langle \nabla f(x) - \nabla f(y) - \frac{\mu}{2}(x-y); x-y \rangle \geq 0$ which is the same

or $\langle \nabla f(x) - \nabla f(y); x-y \rangle \geq \mu \|x-y\|^2$ so

$\langle \nabla f(x) - \nabla f(y); x-y \rangle \geq \mu \|x-y\|^2$ is the same as $f(x)$ is strong convex.

Ex 16

If $f: \mathbb{R}^d \to \mathbb{R}$ is ~~twice~~ $\in C^2(\mathbb{R}^d)$, then

it is convex iff $\mu I \preceq \nabla^2 f(x)$

$g(x) := f(x) - \frac{\mu}{2}\|x\|^2$ then $g(x) \in C^2(\mathbb{R})$

1) If $\nabla^2 f(x) \succeq I\mu$ then $\nabla^2 g(x) \succeq 0$ then $g(x)$ is convex

then $f(x)$ is μ-convex

2) If $f(x)$ is μ-convex then $g(x)$ is convex then

$\nabla^2 g(x) \succeq 0$ then $\nabla^2 f(x) \succeq \mu I$.